

Degree-zero rational cohomology does not decay

Disproof of conjecture 00000001646

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The conjecture asserts that $\dim_{\mathbb{Q}} H^i(\text{Mod}(\Sigma_g); \mathbb{Q})$ tends to zero exponentially as g grows, whenever $i \leq (2g - 2)/3$. Both language versions include degree $i = 0$: no positive-degree or reduced-cohomology restriction is stated. Ordinary group cohomology in this degree gives a counterexample.

Degree-zero calculation for every group. Let G be any group and give $\mathbb{Q}$ the trivial G -action. In the inhomogeneous cochain complex,

$$C^0(G; \mathbb{Q}) = \mathbb{Q}, \quad C^1(G; \mathbb{Q}) = \{f : G \rightarrow \mathbb{Q}\}, \quad (d^0 q)(h) = h \cdot q - q.$$

Since the action is trivial, $d^0 = 0$. Thus every rational number is a zero-cocycle. There are no negative-degree cochains, so the zero-coboundary space is $\text{im}(d^{-1}) = 0$. The actual cohomology quotient is therefore

$$H^0(G; \mathbb{Q}) = \ker(d^0) / \text{im}(d^{-1}) = \mathbb{Q} / 0 \cong \mathbb{Q}, \quad \dim_{\mathbb{Q}} H^0(G; \mathbb{Q}) = 1.$$

This calculation applies to every mapping class group, irrespective of genus, connectedness conventions for surfaces or the choice between standard mapping class group variants.

Failure in the claimed stable range. For every integer $g \geq 1$ we have $0 \leq (2g - 2)/3$. Hence $i = 0$ belongs to the conjecture's stable range at all sufficiently large genera. Nevertheless,

$$\dim_{\mathbb{Q}} H^0(\text{Mod}(\Sigma_g); \mathbb{Q}) = 1 \quad \text{for every } g.$$

The sequence has limit one and cannot have limit zero, much less exponential decay with constant $-\log 2$. Equivalently, the neighborhood $(-\frac{1}{2}, \frac{1}{2})$ of zero contains none of its terms.

Scope. The introductory sentence mentions l^2 -Betti numbers, whereas the claimed quantitative formula explicitly uses ordinary rational cohomology $H^i(-; \mathbb{Q})$. These are different invariants. The counterexample refutes the displayed ordinary-cohomology claim as written. It neither asserts nor refutes any general l^2 -vanishing theorem. Replacing H^0 by reduced cohomology or explicitly restricting to positive degrees would change the statement and would require a separate analysis.

Lean certificate. The project constructs trivial-action cochains, the degree-zero differential, its cocycle kernel, negative-degree zero cochains and the boundary image. An explicit linear equivalence identifies the quotient with $\mathbb{Q}$, proving dimension one for every group. It verifies stable-range inclusion and proves nonconvergence to zero for every group family, including when dimensions are only specified on the stable tail.